Current Progress and Next Steps

Team 43

Completed Work

- Integrated SUMO, OMNeT++ and Veins co-simulation environment.
- Implemented V2V/V2I communication and OBU-based heuristic filtering.
- Integrated RSU-based ML detection and trust management.
- Embedded trained ML models (Random Forest and Logistic Regression) as native C++ predictors in the RSU for real-time edge inference.
- Implemented approximately 15 attack scenarios aligned with VeReMi categories.
- Completed 14-feature extraction and CSV logging pipeline.
- Added trust-score updates and live detection visualization.
- Implemented simulation and trust-state checkpointing.
- Expanded to a multi-grid 4-RSU scenario (rsu[0..3]) across two interconnected intersections.
- Implemented inter-RSU (I2I) communication backbone for periodic trust synchronization and urgent threat alerts.
- Implemented multi-tier Fog layer (fog[0..1]) with regional ledger aggregation, conflict resolution (Bayesian consensus and safety-critical override), and inter-Fog synchronization.
- Integrated SDN-based adaptive forwarding with Dijkstra trust-cost routing and 3-class priority QoS scheduling.
- Built an interactive web dashboard for real-time model evaluation and data inspection.

Remaining Work

- Test scalability with more RSUs and larger network scenarios (Target: 18th October).
- Evaluate accuracy, precision, recall, F1-score, false-positive rate, latency, and packet delivery ratio across diverse traffic densities.